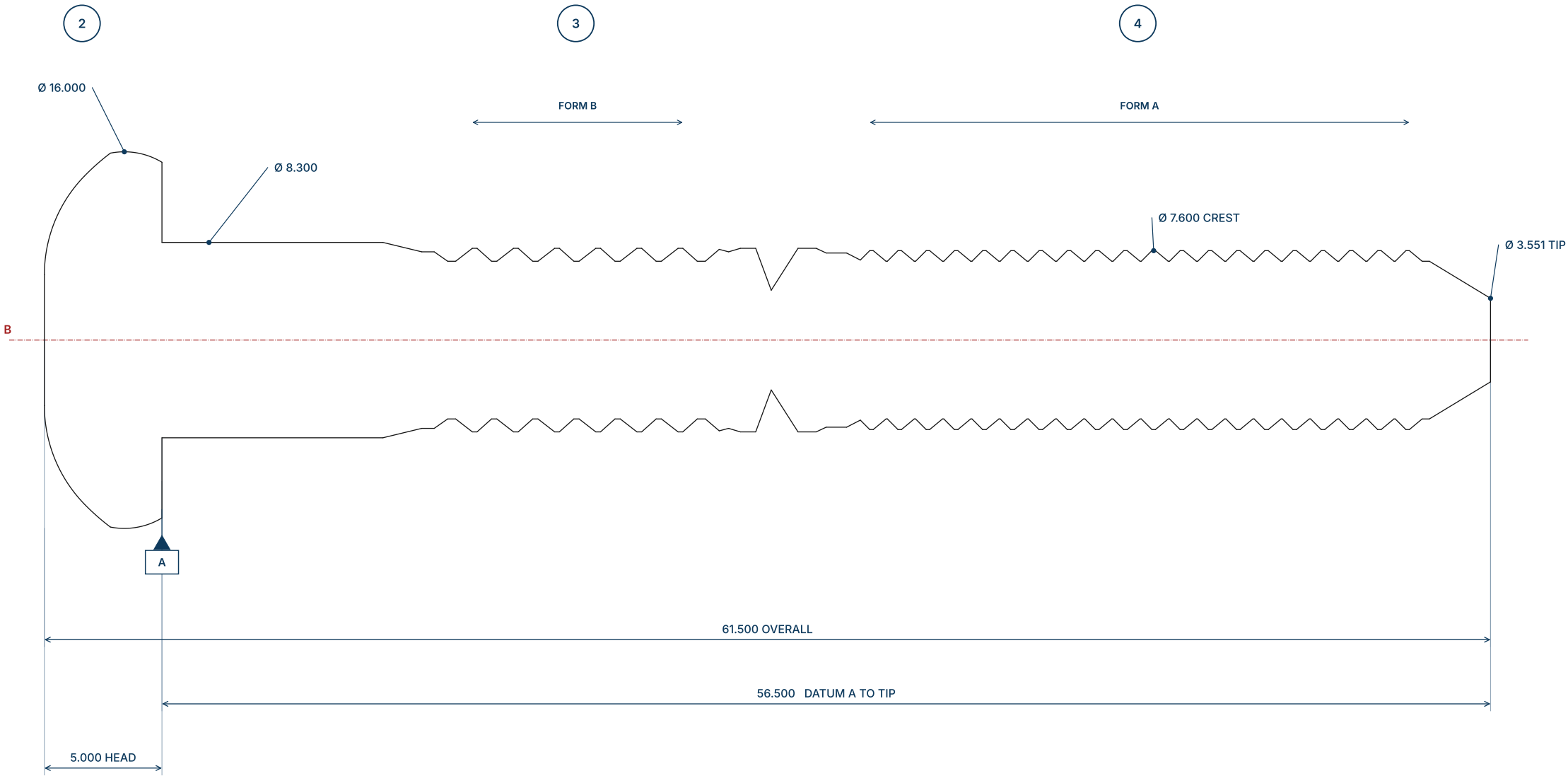


PICTORIAL — NOT TO SCALE HEAD AT LEFT



SCALE 4.4 : 1 AXIS HORIZONTAL HEAD AT LEFT FULL PROFILE

Solid of revolution about datum B. Revolve the +R profile.

Sizes are CAD nominals from 2 MEDIUM NUT.STEP (SolidWorks 2023, 2024-10-24).

No tolerance, material, or finish is stored in the model.

Graphic sizes are rounded to 0.001 mm. Sheets 6 and 7 list every segment to 0.00001 mm.

Positive Z runs from datum A toward the small end.

Volume 3188.318 mm³ area 2052.818 mm² centre of mass Z 19.144 on the axis.

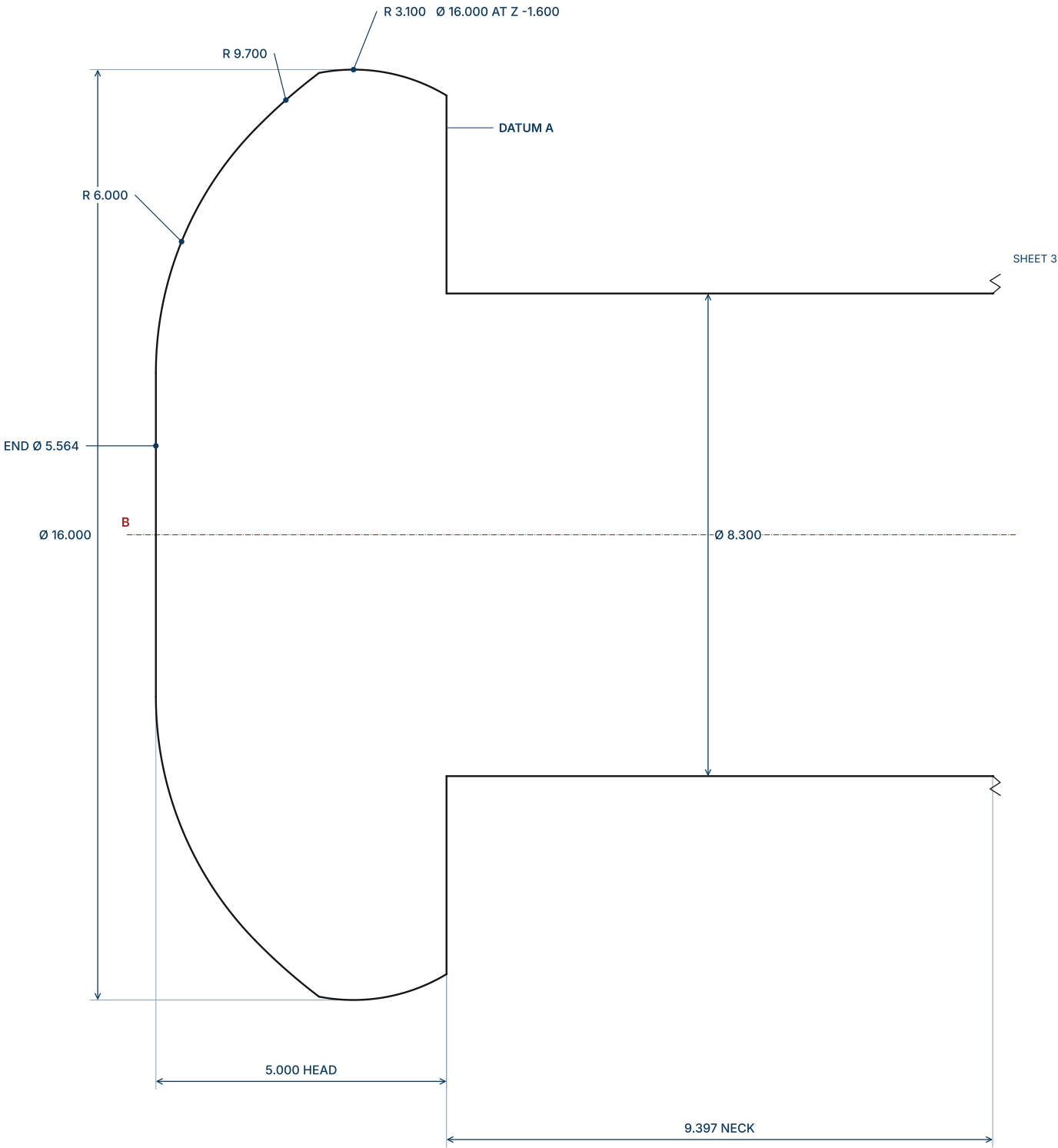
Balloons 2, 3 and 4 mark the detail sheets.

Form B: 6 crests, pitch 1.750, crest Ø 7.800. Form A: 20 crests, pitch 1.200, crest Ø 7.600.



PART	2 MEDIUM NUT	SHEET	1 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0		DATUM B AXIS	

DETAIL 2 SCALE 10 : 1 FULL PROFILE



POINT	Z	RADIUS	Ø	NOTE
HEAD FACE	-5.00000	0.00000	0.000	ON THE AXIS
HEAD RIM	-5.00000	2.78215	5.56430	R6 MEETS THE FACE
CROWN	-1.60000	8.00000	16.00000	LARGEST DIAMETER
DATUM A OUT	0.00000	7.55518	15.11037	SHOULDER OUTER
DATUM A IN	0.00000	4.15000	8.30000	SHOULDER INNER
NECK END	9.39660	4.15000	8.30000	13.60° TAPER STARTS

AXIAL-SECTION ARCS

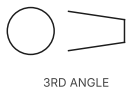
R	CENTER R	CENTER Z	Z0	Z1	ARC LEN
6.00000	2.78215	1.00000	-5.00000	-3.29730	4.63459
9.70000	0.20000	3.65000	-3.29730	-2.19290	1.47353
3.10000	4.90000	-1.60000	-2.19290	0.00000	2.27778

The R3.100 arc carries Ø 16.000.

R6.000 is tangent to the head end face.

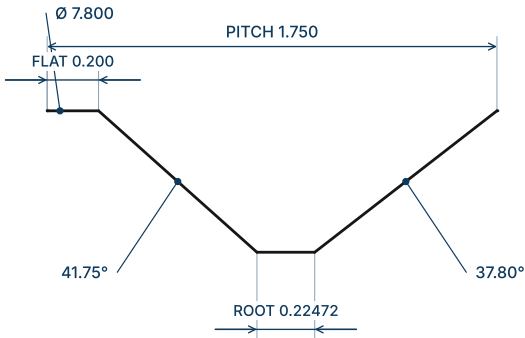
Datum A is the annular face from Ø 8.300 to the head arc.

The neck is a true cylinder. The 13.60° taper is on sheet 3.

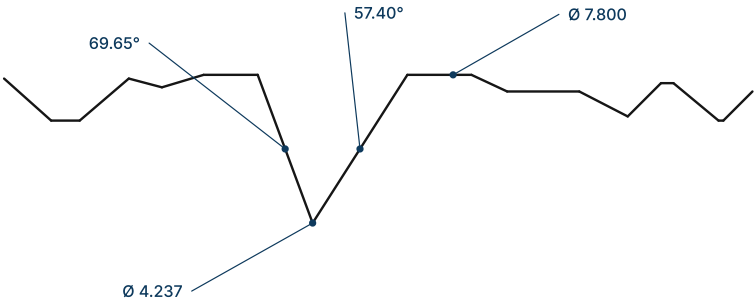


PART	2 MEDIUM NUT	SHEET	2 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A SHOULDERS Z = 0		DATUM B AXIS	

DETAIL 3 ONE PITCH ENLARGED GROOVE AT TRUE RADIAL SCALE ANGLES FROM THE AXIS



FORM B TOOTH SCALE 34 : 1 MATERIAL BELOW THE LINE Ø 6.700 ROOT 6 CRESTS



GROOVE BETWEEN THE FORMS SCALE 11 : 1 HALF PROFILE REMAINING TRANSITIONS ON SHEET 5

#	Z0	Z1	PITCH
1	13.20000	13.40000	—
2	14.95000	15.15000	1.75000
3	16.70000	16.90000	1.75000
4	18.45000	18.65000	1.75000
5	20.20000	20.40000	1.75000
6	21.95000	22.15000	1.75000

FORM B DEFINITION

ITEM	VALUE	NOTE
CREST Ø	7.800	FLAT 0.20000
ROOT Ø	6.700	FLAT 0.22472 ×5
FLANKS	41.75° / 37.80°	LEAVING / APPROACHING, +Z
PITCH	1.75000	CREST START TO CREST START
COUNT	6	Z 13.20000 TO 22.15000
RUNOUT ROOT	0.34323	Z 22.76622 Ø 6.700

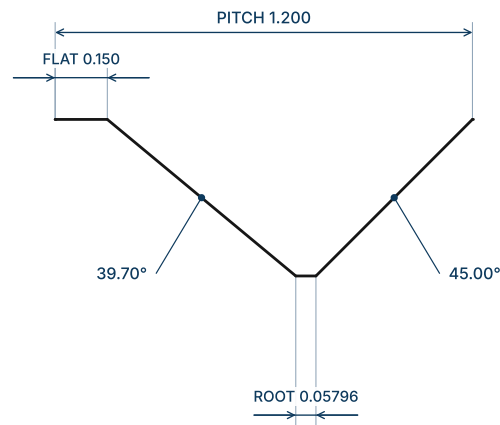
NECK TO FIRST FORM B CREST

FEATURE	Z0	Z1	Ø START	Ø END
13.60°	9.39660	11.05000	8.30000	7.50000
CYL L 0.52874	11.05000	11.57874	7.50000	7.50000
35.00°	11.57874	12.15000	7.50000	6.70000
CYL L 0.34094	12.15000	12.49094	6.70000	6.70000
37.80°	12.49094	13.20000	6.70000	7.80000

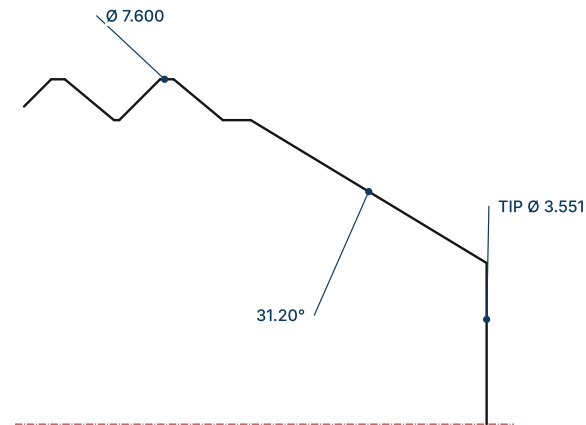


PART	2 MEDIUM NUT	SHEET	3 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0	DATUM B AXIS		

DETAIL 4 REGULAR CREST ENLARGED TIP AT TRUE RADIAL SCALE



CRESTS 2~20 SCALE 46 : 1 Ø 7.600 CREST Ø 6.700 ROOT 20 CRESTS AT PITCH 1.200

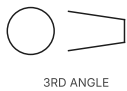


TIP SCALE 12 : 1 HALF PROFILE

#	Z0	Z1	PITCH	#	Z0	Z1	PITCH
1	30.10000	30.25000	—	11	42.10000	42.25000	1.20000
2	31.30000	31.45000	1.20000	12	43.30000	43.45000	1.20000
3	32.50000	32.65000	1.20000	13	44.50000	44.65000	1.20000
4	33.70000	33.85000	1.20000	14	45.70000	45.85000	1.20000
5	34.90000	35.05000	1.20000	15	46.90000	47.05000	1.20000
6	36.10000	36.25000	1.20000	16	48.10000	48.25000	1.20000
7	37.30000	37.45000	1.20000	17	49.30000	49.45000	1.20000
8	38.50000	38.65000	1.20000	18	50.50000	50.65000	1.20000
9	39.70000	39.85000	1.20000	19	51.70000	51.85000	1.20000
10	40.90000	41.05000	1.20000	20	52.90000	53.05000	1.20000

FORM A NOTES

ITEM	VALUE	NOTE
CREST / ROOT	Ø 7.600 / Ø 6.700	FLAT 0.15000 / 0.05796
FLANKS	39.70° / 45.00°	LEAVING / APPROACHING A CREST, +Z
CREST 1 HEAD SIDE	45.00°	STOPS AT Ø 6.800
TIP ROOT	0.30797	Z 53.59203 BEFORE THE 31.20°
TIP FACE	56.50000	Ø 3.55077 TO THE AXIS



3RD ANGLE

PART	2 MEDIUM NUT	SHEET	4 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A SHOULDERS Z = 0		DATUM B AXIS	

CONTROLLING SIZES. Z FROM DATUM A, POSITIVE TOWARD THE SMALL END. ANGLES FROM THE AXIS. MODEL VALUES TO 0.00001 mm.

1 OVERALL

ITEM	MODEL	NOTE
OVERALL	61.50000	HEAD FACE TO TIP FACE
HEAD FACE Z	-5.00000	Ø 5.56430
DATUM A	0.00000	Ø 8.30000 TO 15.11037
LARGEST Ø	16.00000	AT Z -1.60000
TIP FACE Z	56.50000	Ø 3.55077
VOLUME	3188.31805	mm ³ SOLID, NO BORE
SURFACE	2052.81837	mm ²
CON Z	19.14411	ON THE AXIS

2 ARCS

R	CTR R	CTR Z	Z0	Z1
6.00000	2.78215	1.00000	-5.00000	-3.29730
9.70000	0.20000	3.65000	-3.29730	-2.19290
3.10000	4.90000	-1.60000	-2.19290	0.00000

3 NECK

ITEM	Z0	Z1	SIZE
CYLINDER	0.00000	9.39660	Ø 8.30000 L 9.39660

4 AXIS IN THE STEP FILE

ITEM	X	Y	Z
DIRECTION	0.000000	0.000000	-1.000000
POINT ON AXIS	0.10000	0.00000	-5.00000

The point is the head-end centre. World +Z runs from the head toward the tip.

5 EVERY CONTROLLING FEATURE

ITEM	Z0	Z1	SIZE
ARC R6.000	-5.00000	-3.29730	CTR (2.78215, 1.00000)
ARC R9.700	-3.29730	-2.19290	CTR (0.20000, 3.65000)
ARC R3.100	-2.19290	0.00000	CTR (4.90000, -1.60000)
NECK	0.00000	9.39660	Ø 8.30000
TAP 13.60°	9.39660	11.05000	Ø 8.30000 → 7.50000
CYL	11.05000	11.57874	Ø 7.50000 L 0.52874
TAP 35.00°	11.57874	12.15000	Ø 7.50000 → 6.70000
CYL	12.15000	12.49094	Ø 6.70000 L 0.34094
TAP 37.80°	12.49094	13.20000	Ø 6.70000 → 7.80000
B CREST 1	13.20000	13.40000	Ø 7.80000 FLAT 0.20000
B CREST 6	21.95000	22.15000	PITCH 1.75000
TAP 41.75°	22.15000	22.76622	Ø 7.80000 → 6.70000
CYL	22.76622	23.10945	Ø 6.70000 L 0.34323
TAP 40.55°	23.10945	23.70000	Ø 6.70000 → 7.71054
TAP 14.80°	23.70000	24.09843	Ø 7.71054 → 7.50000
TAP 16.65°	24.09843	24.60000	Ø 7.50000 → 7.80000
CYL	24.60000	25.25000	Ø 7.80000 L 0.65000
TAP 69.65°	25.25000	25.91073	Ø 7.80000 → 4.23716
TAP 57.40°	25.91073	27.05000	Ø 4.23716 → 7.80000
CYL	27.05000	27.82045	Ø 7.80000 L 0.77045
TAP 24.97°	27.82045	28.25000	Ø 7.80000 → 7.40000
CYL	28.25000	29.11750	Ø 7.40000 L 0.86750
TAP 27.25°	29.11750	29.69999	Ø 7.40000 → 6.80000
TAP 45.00°	29.69999	30.10000	Ø 6.80000 → 7.60000
A CREST 1	30.10000	30.25000	Ø 7.60000 FLAT 0.15000
A CREST 20	52.90000	53.05000	PITCH 1.20000
TIP 31.20°	53.90000	56.50000	Ø 6.70000 → 3.55077
TIP FACE	56.50000	56.50000	Ø 3.55077 TO THE AXIS
ROOT ×1	12.15000	12.49094	FLAT 0.34094 Ø 6.70000
ROOT ×5	14.01622	21.24094	FLAT 0.22472 Ø 6.70000
ROOT ×1	22.76622	23.10945	FLAT 0.34323 Ø 6.70000
ROOT ×19	30.79203	52.44998	FLAT 0.05796 Ø 6.70000
ROOT ×1	53.59203	53.90000	FLAT 0.30797 Ø 6.70000



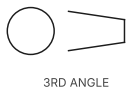
PART	2 MEDIUM NUT	SHEET	5 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A · SHOULDER Z = 0		DATUM B · AXIS	

ORDERED +R PROFILE, HEAD FACE TO TIP. REVOLVE ABOUT DATUM B. MILLIMETRES TO 0.00001. ANGLE IS FROM THE AXIS.

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
001	FACE	-5.00000	0.00000	-5.00000	2.78215	2.78215	90.00	
002	ARC	-5.00000	2.78215	-3.29730	6.96942	4.63459		R6.00000
003	ARC	-3.29730	6.96942	-2.19290	7.94277	1.47353		R9.70000
004	ARC	-2.19290	7.94277	0.00000	7.55518	2.27778		R3.10000
005	FACE	0.00000	7.55518	0.00000	4.15000	3.40518	90.00	
006	CYL	0.00000	4.15000	9.39660	4.15000	9.39660	0.00	
007	TAPER	9.39660	4.15000	11.05000	3.75000	1.70110	13.60	
008	CYL	11.05000	3.75000	11.57874	3.75000	0.52874	0.00	
009	TAPER	11.57874	3.75000	12.15000	3.35000	0.69738	35.00	
010	CYL	12.15000	3.35000	12.49094	3.35000	0.34094	0.00	
011	TAPER	12.49094	3.35000	13.20000	3.90000	0.89736	37.80	
012	CYL	13.20000	3.90000	13.40000	3.90000	0.20000	0.00	
013	TAPER	13.40000	3.90000	14.01622	3.35000	0.82597	41.75	
014	CYL	14.01622	3.35000	14.24094	3.35000	0.22472	0.00	
015	TAPER	14.24094	3.35000	14.95000	3.90000	0.89736	37.80	
016	CYL	14.95000	3.90000	15.15000	3.90000	0.20000	0.00	
017	TAPER	15.15000	3.90000	15.76622	3.35000	0.82597	41.75	
018	CYL	15.76622	3.35000	15.99094	3.35000	0.22472	0.00	
019	TAPER	15.99094	3.35000	16.70000	3.90000	0.89736	37.80	
020	CYL	16.70000	3.90000	16.90000	3.90000	0.20000	0.00	
021	TAPER	16.90000	3.90000	17.51622	3.35000	0.82597	41.75	
022	CYL	17.51622	3.35000	17.74094	3.35000	0.22472	0.00	
023	TAPER	17.74094	3.35000	18.45000	3.90000	0.89736	37.80	
024	CYL	18.45000	3.90000	18.65000	3.90000	0.20000	0.00	
025	TAPER	18.65000	3.90000	19.26622	3.35000	0.82597	41.75	
026	CYL	19.26622	3.35000	19.49094	3.35000	0.22472	0.00	
027	TAPER	19.49094	3.35000	20.20000	3.90000	0.89736	37.80	
028	CYL	20.20000	3.90000	20.40000	3.90000	0.20000	0.00	
029	TAPER	20.40000	3.90000	21.01622	3.35000	0.82597	41.75	
030	CYL	21.01622	3.35000	21.24094	3.35000	0.22472	0.00	
031	TAPER	21.24094	3.35000	21.95000	3.90000	0.89736	37.80	
032	CYL	21.95000	3.90000	22.15000	3.90000	0.20000	0.00	
033	TAPER	22.15000	3.90000	22.76622	3.35000	0.82597	41.75	
034	CYL	22.76622	3.35000	23.10945	3.35000	0.34323	0.00	
035	TAPER	23.10945	3.35000	23.70000	3.85527	0.77721	40.55	
036	TAPER	23.70000	3.85527	24.09843	3.75000	0.41211	14.80	
037	TAPER	24.09843	3.75000	24.60000	3.90000	0.52352	16.65	
038	CYL	24.60000	3.90000	25.25000	3.90000	0.65000	0.00	
039	TAPER	25.25000	3.90000	25.91073	2.11858	1.90001	69.65	
040	TAPER	25.91073	2.11858	27.05000	3.90000	2.11456	57.40	
041	CYL	27.05000	3.90000	27.82045	3.90000	0.77045	0.00	
042	TAPER	27.82045	3.90000	28.25000	3.70000	0.47383	24.97	
043	CYL	28.25000	3.70000	29.11750	3.70000	0.86750	0.00	
044	TAPER	29.11750	3.70000	29.69999	3.40000	0.65520	27.25	
045	TAPER	29.69999	3.40000	30.10000	3.80000	0.56570	45.00	
046	CYL	30.10000	3.80000	30.25000	3.80000	0.15000	0.00	
047	TAPER	30.25000	3.80000	30.79203	3.35000	0.70448	39.70	
048	CYL	30.79203	3.35000	30.84998	3.35000	0.05796	0.00	

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
049	TAPER	30.84998	3.35000	31.30000	3.80000	0.63641	45.00	
050	CYL	31.30000	3.80000	31.45000	3.80000	0.15000	0.00	
051	TAPER	31.45000	3.80000	31.99203	3.35000	0.70448	39.70	
052	CYL	31.99203	3.35000	32.04998	3.35000	0.05796	0.00	
053	TAPER	32.04998	3.35000	32.50000	3.80000	0.63641	45.00	
054	CYL	32.50000	3.80000	32.65000	3.80000	0.15000	0.00	
055	TAPER	32.65000	3.80000	33.19203	3.35000	0.70448	39.70	
056	CYL	33.19203	3.35000	33.24998	3.35000	0.05796	0.00	
057	TAPER	33.24998	3.35000	33.70000	3.80000	0.63641	45.00	
058	CYL	33.70000	3.80000	33.85000	3.80000	0.15000	0.00	
059	TAPER	33.85000	3.80000	34.39203	3.35000	0.70448	39.70	
060	CYL	34.39203	3.35000	34.44998	3.35000	0.05796	0.00	
061	TAPER	34.44998	3.35000	34.90000	3.80000	0.63641	45.00	
062	CYL	34.90000	3.80000	35.05000	3.80000	0.15000	0.00	
063	TAPER	35.05000	3.80000	35.59203	3.35000	0.70448	39.70	
064	CYL	35.59203	3.35000	35.64998	3.35000	0.05796	0.00	
065	TAPER	35.64998	3.35000	36.10000	3.80000	0.63641	45.00	
066	CYL	36.10000	3.80000	36.25000	3.80000	0.15000	0.00	
067	TAPER	36.25000	3.80000	36.79203	3.35000	0.70448	39.70	
068	CYL	36.79203	3.35000	36.84998	3.35000	0.05796	0.00	
069	TAPER	36.84998	3.35000	37.30000	3.80000	0.63641	45.00	
070	CYL	37.30000	3.80000	37.45000	3.80000	0.15000	0.00	
071	TAPER	37.45000	3.80000	37.99203	3.35000	0.70448	39.70	
072	CYL	37.99203	3.35000	38.04998	3.35000	0.05796	0.00	
073	TAPER	38.04998	3.35000	38.50000	3.80000	0.63641	45.00	
074	CYL	38.50000	3.80000	38.65000	3.80000	0.15000	0.00	
075	TAPER	38.65000	3.80000	39.19203	3.35000	0.70448	39.70	
076	CYL	39.19203	3.35000	39.24998	3.35000	0.05796	0.00	
077	TAPER	39.24998	3.35000	39.70000	3.80000	0.63641	45.00	
078	CYL	39.70000	3.80000	39.85000	3.80000	0.15000	0.00	
079	TAPER	39.85000	3.80000	40.39203	3.35000	0.70448	39.70	
080	CYL	40.39203	3.35000	40.44998	3.35000	0.05796	0.00	
081	TAPER	40.44998	3.35000	40.90000	3.80000	0.63641	45.00	
082	CYL	40.90000	3.80000	41.05000	3.80000	0.15000	0.00	
083	TAPER	41.05000	3.80000	41.59203	3.35000	0.70448	39.70	
084	CYL	41.59203	3.35000	41.64998	3.35000	0.05796	0.00	
085	TAPER	41.64998	3.35000	42.10000	3.80000	0.63641	45.00	
086	CYL	42.10000	3.80000	42.25000	3.80000	0.15000	0.00	
087	TAPER	42.25000	3.80000	42.79203	3.35000	0.70448	39.70	
088	CYL	42.79203	3.35000	42.84998	3.35000	0.05796	0.00	
089	TAPER	42.84998	3.35000	43.30000	3.80000	0.63641	45.00	
090	CYL	43.30000	3.80000	43.45000	3.80000	0.15000	0.00	
091	TAPER	43.45000	3.80000	43.99203	3.35000	0.70448	39.70	
092	CYL	43.99203	3.35000	44.04998	3.35000	0.05796	0.00	
093	TAPER	44.04998	3.35000	44.50000	3.80000	0.63641	45.00	
094	CYL	44.50000	3.80000	44.65000	3.80000	0.15000	0.00	
095	TAPER	44.65000	3.80000	45.19203	3.35000	0.70448	39.70	
096	CYL	45.19203	3.35000	45.24998	3.35000	0.05796	0.00	

SEGMENTS 001-096 OF 126



PART	2 MEDIUM NUT	SHEET	6 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A	SHOULDER Z = 0	DATUM B	AXIS

ORDERED +R PROFILE, HEAD FACE TO TIP. REVOLVE ABOUT DATUM B. MILLIMETRES TO 0.00001. ANGLE IS FROM THE AXIS.

#	KIND	Z1	R1	Z2	R2	LENGTH	ANGLE	ARC
097	TAPER	45.24998	3.35000	45.70000	3.80000	0.63641	45.00	
098	CYL	45.70000	3.80000	45.85000	3.80000	0.15000	0.00	
099	TAPER	45.85000	3.80000	46.39203	3.35000	0.70448	39.70	
100	CYL	46.39203	3.35000	46.44998	3.35000	0.05796	0.00	
101	TAPER	46.44998	3.35000	46.90000	3.80000	0.63641	45.00	
102	CYL	46.90000	3.80000	47.05000	3.80000	0.15000	0.00	
103	TAPER	47.05000	3.80000	47.59203	3.35000	0.70448	39.70	
104	CYL	47.59203	3.35000	47.64998	3.35000	0.05796	0.00	
105	TAPER	47.64998	3.35000	48.10000	3.80000	0.63641	45.00	
106	CYL	48.10000	3.80000	48.25000	3.80000	0.15000	0.00	
107	TAPER	48.25000	3.80000	48.79203	3.35000	0.70448	39.70	
108	CYL	48.79203	3.35000	48.84998	3.35000	0.05796	0.00	
109	TAPER	48.84998	3.35000	49.30000	3.80000	0.63641	45.00	
110	CYL	49.30000	3.80000	49.45000	3.80000	0.15000	0.00	
111	TAPER	49.45000	3.80000	49.99203	3.35000	0.70448	39.70	
112	CYL	49.99203	3.35000	50.04998	3.35000	0.05796	0.00	
113	TAPER	50.04998	3.35000	50.50000	3.80000	0.63641	45.00	
114	CYL	50.50000	3.80000	50.65000	3.80000	0.15000	0.00	
115	TAPER	50.65000	3.80000	51.19203	3.35000	0.70448	39.70	
116	CYL	51.19203	3.35000	51.24998	3.35000	0.05796	0.00	
117	TAPER	51.24998	3.35000	51.70000	3.80000	0.63641	45.00	
118	CYL	51.70000	3.80000	51.85000	3.80000	0.15000	0.00	
119	TAPER	51.85000	3.80000	52.39203	3.35000	0.70448	39.70	
120	CYL	52.39203	3.35000	52.44998	3.35000	0.05796	0.00	
121	TAPER	52.44998	3.35000	52.90000	3.80000	0.63641	45.00	
122	CYL	52.90000	3.80000	53.05000	3.80000	0.15000	0.00	
123	TAPER	53.05000	3.80000	53.59203	3.35000	0.70448	39.70	
124	CYL	53.59203	3.35000	53.90000	3.35000	0.30797	0.00	
125	TAPER	53.90000	3.35000	56.50000	1.77538	3.03964	31.20	
126	FACE	56.50000	1.77538	56.50000	0.00000	1.77538	90.00	

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PART	2 MEDIUM NUT	SHEET	7 / 7
FILE	2 MEDIUM NUT.STEP	UNITS	MILLIMETRES
DATUM A SHOULDER Z = 0		DATUM B AXIS	